Lecture 10 Discretion vs. Commitment

Myopic optimiser: Optimize period by period

... discretion: period-by-period optimization, without committing to future actions

Dynamic optimiser: Optimize the sum of present and future utilities.

... commitment: central bank can credibly announce and follow a future path. (backward induction)

Recap: Optimal Monetary Policy is strict inflation targeting. $\hat{\pi}_t = 0 \Rightarrow \hat{y}_t = \hat{y}_t^n = \hat{y}_t^e$

However, possible and optimal but not feasible. since central bank can not fully stabilize inflation. Facing short-run tradeoff due to supply shock.

↳ NK with imperfection at the supply side.

Under flexible price: $y_t^n \neq y_t^e \Rightarrow$ short-run tradeoff between quantity and prices.

Move oppositely facing supply shock

Assume at steady state, the inefficiency associated with the flexible price is not there.

$$\bar{y} = \bar{y}^n = \bar{y}^e$$

while exists short-run deviations. $\hat{y}_t \neq \hat{y}_t^n \neq \hat{y}_t^e$

Welfare losses are: $E_0 \{\sum_{t=0}^{\infty} \beta^t (\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2)\}$

Define: Welfare relevant output gap as

$$\hat{x}_t = \hat{y}_t - \hat{y}_t^e = \log y_t - \log \bar{y} - (\log y_t^e - \log \bar{y}^e) = \log y_t - \log y_t^e$$

$$\hat{\pi}_t = \log P_t - \log P_{t-1} \qquad \alpha_x = \frac{\kappa}{\varepsilon}.$$

$$\left.\begin{aligned} \tilde{y}_t &= \hat{y}_t - \hat{y}_t^n = \hat{x}_t + (\hat{y}_t^e - \hat{y}_t^n) \\ \hat{\pi}_t &= \beta E_t \{\pi_{t+1}\} + \kappa \tilde{y}_t \end{aligned}\right\} \Rightarrow \hat{\pi}_t = \beta E_t \{\pi_{t+1}\} + \kappa \hat{x}_t + u_t,$$

where $u_t = \underline{\kappa(\hat{y}_t^e - \hat{y}_t^n)}$

exogenous w.r.t. monetary policy

u_t makes it impossible to attain simultaneously zero inflation and an efficient level of output

u_t is cost-push shock, eg: exogenous changes in desired wage or price markups, labor income tax, minimum wage etc.

Assume u_t follows, $u_t = \rho_u u_{t-1} + \varepsilon_t^u$, $\rho_u \in [0, 1)$

NKPC: $\hat{\pi}_t = \beta E_t \{\hat{\pi}_{t+1}\} + \kappa \hat{x}_t + u_t$

DIS: $\hat{x}_t = -\frac{1}{\sigma}(\hat{i}_t - E_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^e) + E_t\{\hat{x}_{t+1}\}$

$r_t^e = \rho + \sigma E_t \{\Delta y_{t+1}^e\}$ ... efficient interest rate that supports efficient allocation

§ Optimal policy under discretion (period-by-period optimization)

$$\min_{\{\hat{x}_t,\hat{\pi}_t\}} \hat{\pi}_t^2 + \alpha_x \hat{x}_t^2$$

$$\text{s.t. } \hat{\pi}_t = \kappa \hat{x}_t + v_t, \text{ where } v_t = \beta E_t\{\hat{\pi}_{t+1}\} + u_t.$$

u_t: exogenous

v_t: independent of current policy.

$$\mathcal{L} = \hat{\pi}_t^2 + \alpha_x \hat{x}_t^2 - \lambda(\hat{\pi}_t - \kappa\hat{x}_t - v_t)$$

$$\frac{\partial \mathcal{L}}{\partial \hat{\pi}_t} = 0 \Rightarrow 2\hat{\pi}_t = \lambda$$

$$\frac{\partial \mathcal{L}}{\partial \hat{x}_t} = 0 \Rightarrow 2\alpha_x \hat{x}_t = -\lambda\kappa \Rightarrow \hat{x}_t = -\frac{\kappa}{\alpha_x}\hat{\pi}_t,\ \forall t \quad \text{targeting rule}$$

To dampen the rise in inflation due to cost-push shock, $\hat{x}_t < 0 \Rightarrow \hat{y}_t < \hat{y}_t^e$

$\Rightarrow$ drive output below efficiency level, negative output gap $\Rightarrow$ Policy trade-off

$$\hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + \kappa\hat{x}_t + u_t = \beta E_t\{\hat{\pi}_{t+1}\} - \frac{\kappa^2}{\alpha_x}\hat{\pi}_t + u_t$$

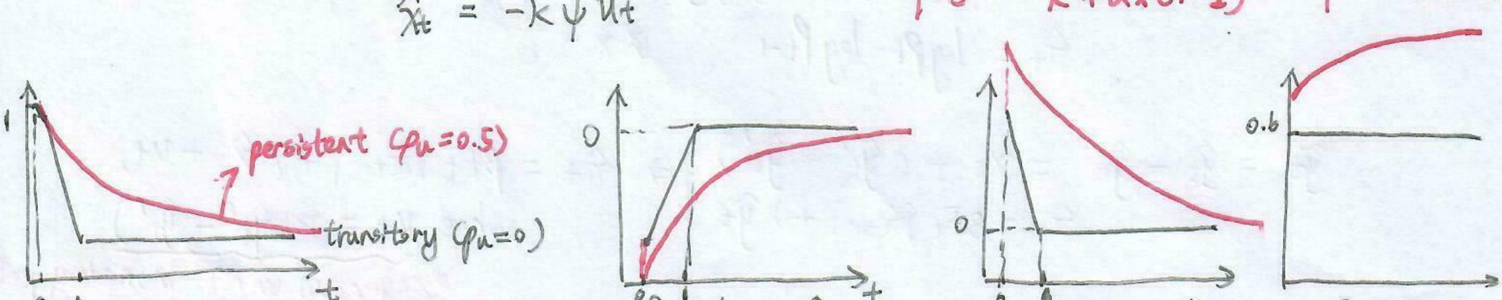
$$\Rightarrow \hat{\pi}_t = \frac{\alpha_x\beta}{\alpha_x+\kappa^2} E_t\{\hat{\pi}_{t+1}\} + \frac{\alpha_x}{\alpha_x+\kappa^2}u_t$$

Iterate forward:
$$\begin{cases} \hat{\pi}_t = \alpha_x \psi u_t, & \text{where } \psi = \dfrac{1}{\kappa^2 + \alpha_x(1-\beta\rho_u)} \\ \hat{x}_t = -\dfrac{\kappa}{\alpha_x}\hat{\pi}_t \Rightarrow \hat{x}_t = -\kappa\psi u_t \end{cases}$$

Central bank let output gap and inflation deviate from targets in proportion to the current value of the cost-push shock:

$$\hat{\pi}_t = \alpha_x \psi u_t$$
$$\hat{x}_t = -\kappa\psi u_t$$

$$\psi_{0.5} = \frac{1}{\kappa^2 + \alpha_x(1-\frac{\beta}{2})} > \psi_0 = \frac{1}{\kappa^2+\alpha_x}$$



Cost push shock. Output gap $\hat{x}_t$ Inflation. $\hat{\pi}_t$ Price level p_t

Optimal to partly accomodate inflation pressure from the cost-push shock by letting inflation rise. Without any policy response, $\kappa = 0$

$$\hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} + u_t \quad \text{Guess that } \hat{\pi}_t = A u_t,\ E_t u_{t+1} = \rho_u u_t$$

$$A u_t = \beta A \rho_u u_t + u_t \Rightarrow A = \frac{1}{1-\beta\rho_u}$$

$\hat{\pi}_t$ rise by less

$$\text{With no output gap response} \Rightarrow \hat{\pi}_t = \frac{1}{1-\beta\rho_u}u_t > \frac{\alpha_x}{\kappa^2+\alpha_x(1-\beta\rho_u)}u_t$$

$\hat{x}_t = -k\psi u_t \Rightarrow k\hat{x}_t = -k^2\psi u_t < 0$ when $u_t > 0$

$$\hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} \boxed{-k^2\psi u_t} + u_t$$

u_t pushes inflation up --- direct effect
Induced recessionary output gap pushes inflation down

Using $E_t \hat{\pi}_{t+1} = \alpha_x \psi \rho_u u_t \Rightarrow \hat{\pi}_t = [\beta \alpha_x \psi \rho_u - k^2\psi + 1] u_t = \alpha_x \psi u_t$

$\hat{\pi}_t = \hat{p}_t - \hat{p}_{t-1} \Rightarrow \hat{p}_t = \hat{p}_{-1} + \sum_{j=0}^{t} \hat{\pi}_j$. since $\hat{\pi}_j = \alpha_x \psi u_j$. since $u_j \to 0$ as $j \to \infty$ $\Rightarrow \hat{\pi}_j \to 0$ as $j \to \infty$

$\Rightarrow \lim_{t\to\infty} \hat{p}_t = \hat{p}_{-1} + \sum_{j=0}^{\infty} \hat{\pi}_j > \hat{p}_{-1}$ ... level price stay permanently higher.

§ Optimal discretionary policy implementation.

optimal policy: choose $(\hat{\pi}_t, \hat{x}_t)$ trade off
(*): implementation, interest rate that decentralizes that solution.

$$\hat{\pi}_t = \alpha_x \psi u_t \qquad \hat{x}_t = -k\psi u_t$$

$$\hat{x}_t = -\frac{1}{\sigma}(\hat{i}_t - E_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^e) + E_t\{\hat{x}_{t+1}\}$$

$$\Rightarrow -k\psi u_t = -\frac{1}{\sigma}(\hat{i}_t - \alpha_x \psi \rho_u u_t - \hat{r}_t^e) - k\psi\rho_u u_t$$

$$\Rightarrow \quad \hat{i}_t = \underbrace{\hat{r}_t^e}_{\text{efficiency rate}} + \underbrace{\psi_i u_t}_{\text{response from cost-push shock}} (*), \text{ where } \psi_i = \psi[\underbrace{k\sigma(1-\rho_u)}_{\text{need to affect current output gap through IS curve}} + \underbrace{\alpha_x \rho_u}_{\text{forward-looking part}}]$$

★ $u_t \uparrow$ cost push shock $\Rightarrow$ inflation $\hat{\pi}_t \uparrow \Rightarrow$ cool demand through $\hat{i}_t \uparrow > \hat{r}_t^e \Rightarrow \hat{x}_t \downarrow \Rightarrow \hat{\pi}_t \downarrow$

but output losses are costly $\Rightarrow$ only fights $\hat{\pi}_t \uparrow$ partially.

How DIS: $\hat{x}_t = -\frac{1}{\sigma}(\hat{i}_t - E_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^e) + E_t\{\hat{x}_{t+1}\}$ is derived?

$C_t^{-\sigma} = \beta(1+i_t) E_t \left[\frac{C_{t+1}^{-\sigma}}{1+\pi_{t+1}}\right] \Rightarrow$ At S.S. $\beta(1+i) = 1$, $\pi = 0$ $\Rightarrow 1 = \beta(1+r) \Rightarrow 1+r = \frac{1}{\beta}$ $\Rightarrow \rho = -\log\beta$.

$$\Rightarrow \hat{c}_t = E_t \hat{c}_{t+1} - \frac{1}{\sigma}[\hat{i}_t - E_t \hat{\pi}_{t+1} - \rho]$$

$$\Rightarrow \hat{y}_t = -\frac{1}{\sigma}(\hat{i}_t - E_t \hat{\pi}_{t+1} - \rho) + E_t \hat{y}_{t+1}$$

ρ → steady state real rate

$$\Rightarrow \hat{x}_t + \hat{y}_t^e = -\frac{1}{\sigma}(\hat{i}_t - E_t\hat{\pi}_{t+1} - \rho) + E_t\hat{x}_{t+1} + E_t \hat{y}_{t+1}^e$$

$$\Rightarrow \hat{x}_t = -\frac{1}{\sigma}(\hat{i}_t - E_t\hat{\pi}_{t+1} - \rho) + E_t\hat{x}_{t+1} + (E_t\hat{y}_{t+1}^e - \hat{y}_t^e)$$

$$\Rightarrow \hat{x}_t = -\frac{1}{\sigma}[\hat{i}_t - E_t\hat{\pi}_{t+1} \underbrace{-\rho - \sigma E_t \Delta \hat{y}_{t+1}^e}_{\hat{r}_t^e}] + E_t\hat{x}_{t+1}$$

$$\Rightarrow \hat{x}_t = -\frac{1}{\sigma}[\hat{i}_t - E_t\hat{\pi}_{t+1} - \hat{r}_t^e] + E_t\hat{x}_{t+1}$$

Consider an inflation targeting rule: $\hat{i}_t = \hat{r}_t^e + \phi_\pi \hat{\pi}_t$ where $\phi_\pi = (1-\rho_u)\frac{\kappa\sigma}{\alpha_x} + \rho_u$.

$$\left.\begin{array}{l} \hat{i}_t = \hat{r}_t^e + \psi_i u_t \\ \hat{\pi}_t = \alpha_x \psi u_t \Rightarrow u_t = \dfrac{\hat{\pi}_t}{\alpha_x \psi} \end{array}\right\} \Rightarrow \hat{i}_t = \hat{r}_t^e + \frac{\psi_i}{\alpha_x \psi}\hat{\pi}_t$$

$$\frac{\psi_i}{\alpha_x \psi} = \frac{[\kappa\sigma(1-\rho_u) + \alpha_x \rho_u]\psi}{\alpha_x \psi} = (1-\rho_u)\frac{\kappa\sigma}{\alpha_x} + \rho_u = \phi_\pi$$

Since a determinate equilibrium exists iff. $\phi_\pi > 1 \Rightarrow \kappa\sigma > \alpha_x$ may not satisfy

$$\hat{i}_t = \hat{r}_t^e + \psi_i u_t + \phi_\pi (\underbrace{\hat{\pi}_t - \alpha_x \psi u_t}_{\text{Inflation deviation from the equilibrium value under policy.}})$$

$$= \hat{r}_t^e + \Theta_i u_t + \phi_\pi \hat{\pi}_t$$

where, $\Theta_i = \psi[\kappa\sigma(1-\rho_u) - \alpha_x(\phi_\pi - \rho_u)]$

Now we have a unique equilibrium, and it is the desired discretionary equilibrium.

1° Under desired equilibrium: $\hat{\pi}_t = \alpha_x \psi u_t$

$\Rightarrow \hat{i}_t = \hat{r}_t^e + \psi_i u_t \Rightarrow$ We do not change the intended discretionary allocation.

2° Central bank reacts more aggressively.

$\hat{i}_t \uparrow$ when $\hat{\pi}_t > \alpha_x \psi u_t$ (off the intended path)

If we have $\phi_\pi > 1$, nominal rate rises more than 1-for-1 with inflation deviations.

$r \uparrow \Rightarrow$ demand $\downarrow \Rightarrow$ inflation $\downarrow \Rightarrow$ eliminate self-fulfilling inflation paths.

Target rule: tell central bank the relationship between target variables to achieve.

eg: $\hat{x}_t = -\frac{\kappa}{\alpha_x}\hat{\pi}_t$ -- tells tradeoff, but still infeasible, since y_t^e unobservable

Instrument rule: tell central bank how to set its instrument

eg: $\hat{i}_t = \underbrace{\hat{r}_t^e + \Theta_i u_t}_{\text{hard to observe and measure}} + \phi_\pi \hat{\pi}_t$

9 Optimal Policy under Commitment.

Central bank able to commit with full credibility to a policy plan.

$$\min_{\{\hat{x}_t, \hat{\pi}_t\}_{t=0}^{\infty}} \frac{1}{2} E_0 \sum_{t=0}^{\infty} \beta^t (\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2)$$

s.t. $\hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + \kappa \hat{x}_t + u_t$

$u_t = \rho_u u_{t-1} + \varepsilon_t^u$

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{1}{2}(\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2) + \gamma_t(\hat{\pi}_t - \kappa \hat{x}_t - \beta \hat{\pi}_{t+1} - u_t)\right]$$

F.O.C. $\frac{\partial \mathcal{L}}{\partial \hat{x}_t}=0 \Rightarrow \alpha_x \hat{x}_t - \kappa \gamma_t = 0 \Rightarrow \gamma_t = \frac{\alpha_x}{\kappa}\hat{x}_t \quad \forall t \qquad \gamma_{-1}=0$

plug in

$\frac{\partial \mathcal{L}}{\partial \hat{\pi}_t}=0 \Rightarrow \hat{\pi}_t + \gamma_t - \gamma_{t-1}=0$

Constraint in period -1 not effective for optimal plan in period 0

$$\Rightarrow \begin{cases} \hat{x}_0 = -\frac{\kappa}{\alpha_x}\hat{\pi}_0 \\ \hat{x}_t = \hat{x}_{t-1} - \frac{\kappa}{\alpha_x}\hat{\pi}_t, \quad t \geq 1 \end{cases} \quad (*) \quad \Rightarrow \hat{x}_t = -\frac{\kappa}{\alpha_x}\check{p}_t \quad \text{where } \check{p}_t = \log P_t - \log P_{-1}$$

↳ price-level targetting

Under discretion, we have purely static: $\hat{x}_t = -\frac{\kappa}{\alpha_x}\hat{\pi}_t$

Now today's policy depends on past policy through $\hat{x}_{t-1}$. history dependent policy making.

Then $\hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + \kappa \hat{x}_t + u_t$

$$\Rightarrow \check{p}_t - \check{p}_{t-1} = \beta E_t\{\check{p}_{t+1} - \check{p}_t\} - \frac{\kappa^2}{\alpha_x}\check{p}_t + u_t$$

$$\Rightarrow \left(1+\beta+\frac{\kappa^2}{\alpha_x}\right)\check{p}_t = \beta E_t\{\check{p}_{t+1}\} + \check{p}_{t-1} + u_t \qquad \text{let } a = \frac{\alpha_x}{(1+\beta)\alpha_x + \kappa^2}$$

$$\Rightarrow \check{p}_t = a\check{p}_{t-1} + a\beta E_t\{\check{p}_{t+1}\} + a u_t, \quad \forall t.$$

Assume $\check{p}_t = \delta \check{p}_{t-1} + \eta u_t$ -- undetermined coefficients.

$$\check{p}_t = \frac{a}{\delta}(\check{p}_t - \eta u_t) + a\beta\delta\check{p}_t + a\beta\eta\rho_u u_t + a u_t$$

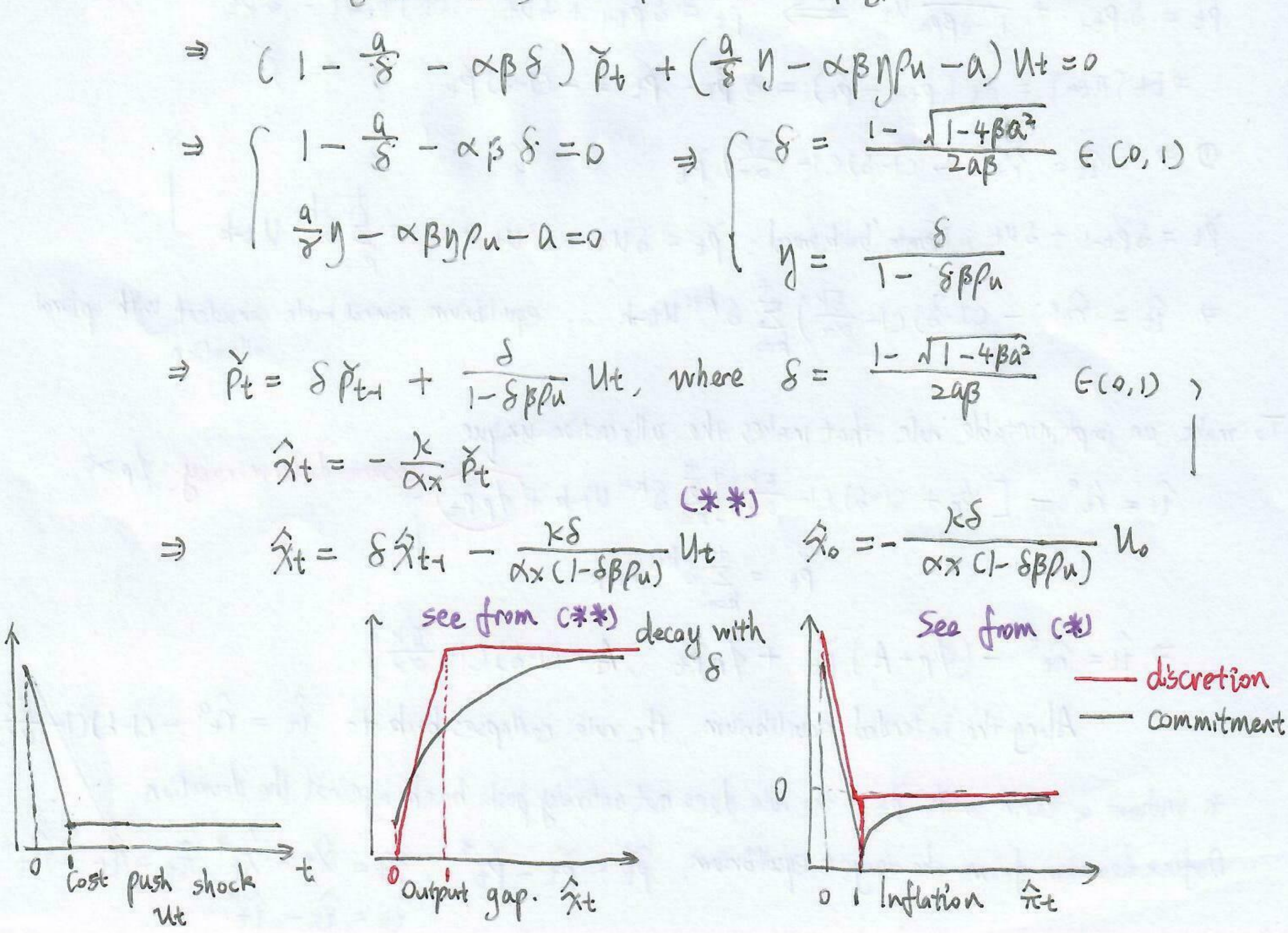
$$\Rightarrow \left(1 - \frac{a}{\delta} - a\beta\delta\right)\check{p}_t + \left(\frac{a}{\delta}\eta - a\beta\eta\rho_u - a\right)u_t = 0$$

$$\Rightarrow \begin{cases} 1 - \frac{a}{\delta} - a\beta\delta = 0 \\ \frac{a}{\delta}\eta - a\beta\eta\rho_u - a = 0 \end{cases} \Rightarrow \begin{cases} \delta = \frac{1-\sqrt{1-4\beta a^2}}{2a\beta} \in (0,1) \\ \eta = \frac{\delta}{1-\delta\beta\rho_u} \end{cases}$$

$$\Rightarrow \check{p}_t = \delta\check{p}_{t-1} + \frac{\delta}{1-\delta\beta\rho_u}u_t, \quad \text{where } \delta = \frac{1-\sqrt{1-4\beta a^2}}{2a\beta} \in (0,1)$$

$$\hat{x}_t = -\frac{\kappa}{\alpha_x}\check{p}_t$$

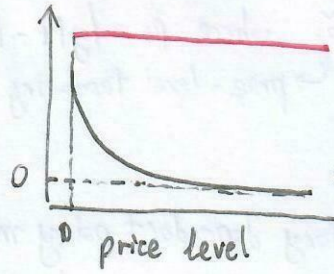
$$\Rightarrow \hat{x}_t = \delta\hat{x}_{t-1} - \frac{\kappa\delta}{\alpha_x(1-\delta\beta\rho_u)}u_t \quad (**) \qquad \hat{x}_0 = -\frac{\kappa\delta}{\alpha_x(1-\delta\beta\rho_u)}u_0$$



$$\hat{\pi}_t = \kappa\hat{x}_t + \kappa\sum_{k=1}^{\infty}\beta^t E_t\{\hat{x}_{t+k}\} + u_t$$

In response to u_t, central bank may lower future output gap with credible promises. Given $\hat{\pi}_t$, current $\hat{x}_t$ may decline less.

* Since cost function is convex, dampening of deviations in period of shock improves welfare



discretion

commitment -- price level with return to S.S.
price-level targeting

Assume transitory shock ($\rho_u = 0$)

$$\hat{x}_t = -\frac{1}{\sigma}(\hat{i}_t - E_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^e) + E_t\{\hat{x}_{t+1}\}$$

$$\Rightarrow \hat{i}_t = \hat{r}_t^e + E_t\{\hat{\pi}_{t+1}\} + \sigma[E_t\{\hat{x}_{t+1}\} - \hat{x}_t]$$

$$\hat{x}_t = -\frac{\kappa}{\alpha_x}\check{p}_t \Rightarrow E_t\{\hat{x}_{t+1}\} - \hat{x}_t = -\frac{\kappa}{\alpha_x}\underbrace{(E_t\check{p}_{t+1} - \check{p}_t)}_{E_t\pi_{t+1}}$$

$$\Rightarrow \hat{i}_t = \hat{r}_t^e + E_t\{\hat{\pi}_{t+1}\} - \sigma\frac{\kappa}{\alpha_x}E_t\{\pi_{t+1}\}$$

$$\Rightarrow \hat{i}_t = \hat{r}_t^e + (1 - \frac{\sigma\kappa}{\alpha_x})E_t\{\hat{\pi}_{t+1}\} \quad ①$$

$$\check{p}_t = \delta\check{p}_{t-1} + \frac{\delta}{1-\delta\beta\rho_u}u_t \xRightarrow{\rho_u = 0} \check{p}_t = \delta\check{p}_{t-1} + \delta u_t \qquad E_t\{\check{p}_{t+1}\} = \delta\check{p}_t$$

$$\Rightarrow E_t\{\pi_{t+1}\} = E_t(\check{p}_{t+1} - \check{p}_t) = \delta\check{p}_t - \check{p}_t = -(1-\delta)\check{p}_t$$

$$① \Rightarrow \hat{i}_t = \hat{r}_t^e - (1-\delta)(1 - \frac{\sigma\kappa}{\alpha_x})\check{p}_t$$

$\check{p}_t = \delta\check{p}_{t-1} + \delta u_t$, iterate backward. $\check{p}_t = \delta u_t + \delta^2 u_t + \cdots = \sum_{k=0}^{t}\delta^{k+1}u_{t-k}$

$\Rightarrow \hat{i}_t = \hat{r}_t^e - (1-\delta)(1 - \frac{\sigma\kappa}{\alpha_x})\sum_{k=0}^{t}\delta^{k+1}u_{t-k}$ ... equilibrium nominal rate consistent with optimal allocation

To make an implementable rule that makes the allocation unique

$$\hat{i}_t = \hat{r}_t^e - [\phi_p + (1-\delta)(1-\frac{\sigma\kappa}{\alpha_x})]\sum_{k=0}^{t}\delta^{k+1}u_{t-k} + \phi_p\check{p}_t$$

ensure determinacy. $\phi_p > 0$

$$\check{p}_t = \sum_{k=0}^{t}\delta^{k+1}u_{t-k}$$

$$\Rightarrow \hat{i}_t = \hat{r}_t^e - (\phi_p + A)\check{p}_t + \phi_p\check{p}_t, \quad A = (1-\delta)(1-\frac{\sigma\kappa}{\alpha_x})$$

Along the intended equilibrium, the rule collapses back to $\hat{i}_t = \hat{r}_t^e - (1-\delta)(1-\frac{\sigma\kappa}{\alpha_x})\check{p}_t$

* Without a term with $\check{p}_t$, the rule does not actively push back against the deviation.

Define deviation from the target equilibrium. $\tilde{p}_t = \check{p}_t - \check{p}_t^*$, $\tilde{x}_t = \hat{x}_t - \hat{x}_t^*$, $\tilde{\pi}_t = \hat{\pi}_t - \hat{\pi}_t^*$, $\tilde{i}_t = \hat{i}_t - \hat{i}_t^*$

shock terms cancel out $\Rightarrow$ $\tilde{i}_t = \phi_p \tilde{p}_t$

$\tilde{\pi}_t = \tilde{p}_t - \tilde{p}_{t-1}$

IS $\Rightarrow$ $\tilde{x}_t = -\frac{1}{\sigma}(\tilde{i}_t - E_t \tilde{\pi}_{t+1}) + E_t \tilde{x}_{t+1}$

$\Rightarrow$ $\tilde{x}_t = -\frac{1}{\sigma}(\phi_p \tilde{p}_t - E_t \tilde{\pi}_{t+1}) + E_t \tilde{x}_{t+1}$ with $\tilde{x}_t = -\frac{\kappa}{\alpha_x}\tilde{p}_t$

$\Rightarrow$ $-\frac{\kappa}{\alpha_x}\tilde{p}_t = -\frac{1}{\sigma}(\phi_p \tilde{p}_t - E_t \tilde{p}_{t+1} + \tilde{p}_t) - \frac{\sigma\kappa}{\alpha_x} E_t \tilde{p}_{t+1}$

$\Rightarrow$ $E_t \tilde{p}_{t+1} = \left(1 + \frac{\phi_p}{1 - \frac{\sigma\kappa}{\alpha_x}}\right)\tilde{p}_t$ if $\phi_p > 0$, coefficient is shifted away from knife-edge case

* The only bounded rational-expectations deviation path is $\tilde{p}_t = 0 \Rightarrow \check{p}_t = \check{p}_t^*$

$\Rightarrow$ If $\phi_p > 0$, $\check{p}_t \uparrow \Rightarrow \hat{i}_t \uparrow$ $\Rightarrow$ reduce demand $\Rightarrow$ negative output gap $\xRightarrow{NKPC}$ $\hat{\pi}_t \downarrow$

$\Rightarrow$ The belief that "prices will drift up" is not self-fullfilling anymore. Policy offsets it.

If $\phi_p = 0$, Central bank does not react to this endogenous price-level drift, beliefs may survive as alternative equilibria

6) Distorted steady state

Permanent supply-side distortion: $\bar{y}^n < \bar{y}^e$

Real imperfections generate a permanent gap between natural and the efficient levels of output.

$$-\frac{U_N}{U_C} = MPN(1-\phi), \quad \phi = 1 - \frac{1}{\mu} \quad \text{output is inefficiently low.}$$

The welfare losses: $E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{1}{2}(\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2) - \Lambda \hat{x}_t\right]$ -- objective, where $\Lambda = \phi\frac{\lambda}{\varepsilon} > 0$

$\hat{x}_t = \log X_t - \log \bar{X}$ -- deviation of the welfare-relevant output gap.

$\log X_t = \log Y_t - \log Y_t^e$, $\boxed{\log \bar{X}} = \log \bar{Y} - \log \bar{Y}^e = \log \bar{Y}^n - \log \bar{Y}^e < 0$

$\hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + \kappa \hat{x}_t + u_t$ -- constraint (NKPC), where $u_t = \kappa(\hat{y}_t^e - \hat{y}_t^n)$ is cost-push shock.

i. Optimal discretionary policy: $\min_{\hat{x}_t, \hat{\pi}_t} \frac{1}{2}(\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2) - \Lambda \hat{x}_t$

s.t. $\hat{\pi}_t = \kappa \hat{x}_t + v_t$, where $v_t = \beta E_t\{\pi_{t+1}\} + u_t$

$\Rightarrow$ F.O.C. $\hat{x}_t = \boxed{\frac{\Lambda}{\alpha_x}} - \frac{\kappa}{\alpha_x}\hat{\pi}_t$. plug into the constraint

Central bank always wants a more expansionary policy.

$\Rightarrow$ $\hat{\pi}_t = \frac{\alpha_x \beta}{\alpha_x + \kappa^2} E_t\{\hat{\pi}_{t+1}\} + \frac{\kappa\Lambda}{\alpha_x + \kappa^2} + \frac{\alpha_x}{\alpha_x + \kappa^2} u_t$ Guess $\hat{\pi}_t = A + B u_t$.

$\Rightarrow$ $A + B u_t = \frac{\alpha_x \beta}{\alpha_x + \kappa^2}(A + B\rho_u u_t) + \frac{\kappa\Lambda}{\alpha_x + \kappa^2} + \frac{\alpha_x}{\alpha_x + \kappa^2} u_t$

$\Rightarrow A = \frac{\kappa\Lambda}{\kappa^2 + \alpha_x(1-\beta)}$ $\quad B = \alpha_x \cdot \psi$, where $\psi = \frac{1}{\kappa^2 + \alpha_x(1-\beta\rho_u)}$

$$\Rightarrow \hat{\pi}_t = \underbrace{\frac{\Lambda\kappa}{\kappa^2 + \alpha_x(1-\beta)}}_{\text{constant} > 0} + \underbrace{\alpha_x \psi u_t}_{\text{usual response}}$$

changed average level of inflation / output gap. shock response no change

inflation bias from distorted S.S.

Inefficient low natural level of output $\Rightarrow$
① positive average inflation. because CB's incentive to push output above its natural S.S. level.
② increases the incentive to keep inflation higher than 0 as Λ increases (inflation bias)

ii. Optimal policy under commitment.

$$\mathcal{L} = E_t \sum_{t=0}^{\infty} \beta^t \left[\frac{1}{2}(\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2) - \Lambda \hat{x}_t + \gamma_t (\hat{\pi}_t - \kappa \hat{x}_t - \beta E_t \hat{\pi}_{t+1} - u_t) \right]$$

F.O.C.: $\frac{\partial \mathcal{L}}{\partial \hat{x}_t} = 0 \Rightarrow \alpha_x \hat{x}_t - \kappa\gamma_t - \Lambda = 0$. ①

$\gamma_{-1} = 0$

$\frac{\partial \mathcal{L}}{\partial \hat{\pi}_t} = 0 \Rightarrow \hat{\pi}_t + \gamma_t - \gamma_{t-1} = 0$ ②

① $\Rightarrow \gamma_t = \frac{\alpha_x \hat{x}_t - \Lambda}{\kappa}$ $\quad \gamma_{t-1} = \frac{\alpha_x \hat{x}_{t-1} - \Lambda}{\kappa}$ plug into ② $\Rightarrow \hat{x}_t = \hat{x}_{t-1} - \frac{\kappa}{\alpha_x}\hat{\pi}_t$

since $\hat{\pi}_t = \check{p}_t - \check{p}_{t-1} \xRightarrow[\text{forward}]{\text{Iterate}} \hat{x}_t = -\frac{\kappa}{\alpha_x}\check{p}_t + \text{constant}$ $\Rightarrow$ constant $= \frac{\Lambda}{\alpha_x}$

① $\Rightarrow \hat{x}_t = \frac{\Lambda}{\alpha_x} - \frac{\kappa}{\alpha_x}\check{p}_t$

Plug into NKPC: $\hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} + \kappa \hat{x}_t + u_t$, $\hat{\pi}_t = \check{p}_t - \check{p}_{t-1}$ $\Rightarrow \check{p}_t = a\check{p}_{t-1} + a\beta E_t \check{p}_{t+1} + \frac{a\kappa\Lambda}{\alpha_x} + a u_t$ where $a = \frac{\alpha_x}{\kappa^2 + \alpha_x(1+\beta)}$

solve with undetermined coefficient: $\check{p}_t = A\check{p}_{t-1} + Bu_t + C$

$\Rightarrow A = \delta = \frac{1 - \sqrt{1-4\beta a^2}}{2a\beta} \in (0,1)$, $B = \frac{\delta}{1-\delta\beta\rho_u}$, $C = \frac{\delta\kappa\Lambda}{1-\delta\beta}$

$$\Rightarrow \check{p}_t = \delta \check{p}_{t-1} + \underbrace{\boxed{\frac{\delta}{1-\delta\beta\rho_u}}}_{\text{shock response not affected}} u_t + \underbrace{\boxed{\frac{\delta\kappa\Lambda}{1-\delta\beta}}}_{\text{stabilization bias}} \quad ☆$$

shifts price-level path upward by a constant $\Rightarrow$ price level converges to a constant and average inflation is 0

$\lim_{T\to\infty} p_T = p_{-1} + \frac{\delta\kappa\Lambda}{(1-\delta\beta)(1-\delta)}$ ... zero average inflation, but positive upward bias above p_{-1}

$\hat{x}_t = \frac{\Lambda}{\alpha_x} - \frac{\kappa}{\alpha_x}\check{p}_t$ and plug in ☆ $\Rightarrow \hat{x}_t = \underbrace{\boxed{\delta \hat{x}_{t-1}}}_{\text{history dependence}} - \underbrace{\boxed{\frac{\kappa\delta}{\alpha_x(1-\delta\beta\rho_u)}}}_{\text{same shock response}} u_t + \underbrace{\boxed{\frac{\Lambda}{\alpha_x}\left[1 - \delta\left(1 + \frac{\kappa^2}{1-\delta\beta}\right)\right]}}_{\text{Constant S.S. deviation}}$

Commitment leads to:

① Output can be raised above its natural level with welfare improvement

② more subdued effects on inflation

③ Commitment avoids the inflation bias under discretionary policy.